

Problem Statement for the Domain: Web Application Security

Problem Statement

Web applications are the backbone of modern digital services such as online banking, e-commerce, healthcare, education, and government portals. However, many web applications remain vulnerable to cyberattacks due to insecure coding practices, misconfigurations, and delayed vulnerability assessments. Common threats such as **SQL Injection (SQLi)**, **Cross-Site Scripting (XSS)**, **Cross-Site Request Forgery (CSRF)**, **broken authentication**, **insecure APIs**, and **security misconfigurations** can lead to unauthorized access, data breaches, financial losses, and service disruptions.

Traditional vulnerability assessment tools often require skilled cybersecurity professionals to configure scans, interpret findings, and prioritize remediation. Small organizations and educational institutions frequently lack the expertise and resources needed to perform regular security assessments.

This project proposes the development of an **AI-powered Single-Agent Web Application Security Assessment System** that autonomously scans web applications, identifies security vulnerabilities, classifies risks based on industry standards (such as the **OWASP Top 10**), recommends mitigation strategies, and generates comprehensive security reports. By leveraging AI and automation, the system reduces manual effort, improves the speed and accuracy of vulnerability detection, and assists developers in building more secure web applications.